External monitor not detected or flickering

Article KB0010039 | Owner: Desktop Support | Subcategory: Monitor / docking station | Applies to: Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840, Windows 11 23H2/24H2, macOS 14/15, USB-C/Thunderbolt docking stations (DisplayPort, HDMI outputs)

Summary

This article helps users and Desktop Support agents diagnose and resolve issues where an external monitor connected to a laptop via a docking station is not detected, flickers, or shows an incorrect resolution.

Symptoms

- External monitor remains black or shows "No signal"
- Monitor flickers intermittently or displays artifacts
- Second screen not listed in Display Settings
- Resolution stuck at low values (e.g., 800x600) or refresh rate incorrectly set
- Error message in Windows: "Unable to detect display" or macOS: "No display attached"

Common causes

- Loose or damaged DisplayPort/HDMI cable
- Outdated docking-station firmware
- Missing or corrupted graphics/display driver
- Incorrect refresh rate or resolution selected in OS display settings
- Power-delivery issue causing the dock to reset
- Conflicting multiple display adapters (e.g., Intel UHD + NVIDIA GPU)

Initial Checks

Before proceeding with troubleshooting, verify the basics: ensure the docking station is receiving power (LED indicator on), the laptop is awake and logged in, and the external monitor is powered on and set to the correct input source (DP/HDMI). Disconnect any other USB-C/Thunderbolt peripherals that might interfere with the dock's bandwidth.

Self-Help Procedure: Basic Troubleshooting

Follow these steps to resolve most detection or flickering problems. If the issue persists after completing the procedure, gather the information noted in the 'Information to Collect' section and contact the Desktop Support team via the IT self-service portal, phone extension 4357, or the Teams "IT Help" channel.

1. 1. Power cycle the docking station: unplug the dock from power and from the laptop, wait 10 seconds, then reconnect power first, followed by the laptop cable.
2. 2. Verify the video cable: disconnect the DisplayPort/HDMI cable from both the dock and the monitor, inspect for bent pins or visible damage, re-seat firmly, or replace with a known-good cable.
3. 3. Detect the monitor manually:
   - Windows: Right-click desktop -> Display Settings -> click "Detect".
   - macOS: Apple menu -> System Settings -> Displays -> hold Option and click "Detect Displays".
4. 4. Set the recommended resolution and refresh rate:
   - Windows: Under Display Settings, select the external monitor, choose the resolution marked "Recommended", and set the refresh rate to 60?Hz (or the monitor's native rate).
   - macOS: Select the monitor, choose "Default for display" or manually pick the native resolution and refresh rate.
5. 5. Update the graphics driver:
   - Windows: Open Device Manager -> Expand "Display adapters", right-click your GPU -> "Update driver" -> "Search automatically for drivers".
   - macOS: Ensure you are on the latest system update via Software Update.
6. 6. Check docking-station firmware (if applicable):
   - Visit the dock manufacturer's support site, download the latest firmware utility, and run it while the dock is connected to the laptop.
   - For Dell WD19/TB16, use Dell Command | Update; for Lenovo ThinkPad USB-C Dock Gen 2, use Lenovo Vantage;

for HP docking stations, use HP Support Assistant.
7. 7. Test with an alternative port or monitor:
- If the dock offers multiple DP/HDMI outputs, move the cable to another port.
- If possible, connect a different monitor to isolate whether the issue lies with the monitor or the dock/cable.

Advanced Checks (Cable, Dock Firmware, Drivers)

If the basic self-help steps do not restore the display, perform these additional checks. They require administrative privileges on Windows or the ability to install software on macOS.

1. 1. Capture EDID information:
- Windows: Use the free tool "MonInfo" or "CRU" to read the monitor's EDID from each port; compare values to ensure the dock is passing correct timing data.
- macOS: Hold Option and click Apple menu -> System Information -> Graphics/Displays to see the detected EDID.
2. 2. Verify dock power delivery:
- Ensure the laptop is charging when docked; a failing power-delivery circuit can cause the dock to reset the video subsystem intermittently.
3. 3. Examine event logs for GPU errors:
- Windows: Open Event Viewer -> Windows Logs -> System, filter by Source "nvlddmkm" (NVIDIA) or "dxgmms1" (Intel/AMD) for timestamps coinciding with the flicker.
- macOS: Console.app -> look for "IOGPU" or "WindowServer" messages.
4. 4. Roll back or clean-install the graphics driver:
- Windows: In Device Manager, choose "Uninstall device" and tick "Delete the driver software for this device", then reboot and let Windows reinstall or install the latest driver from the vendor's site.
- macOS: Boot into Safe Mode (Shift at startup) to test if the issue persists; if not, a third-party kernel extension may be interfering.

Service Desk Procedure

Use this workflow when a user has escalated the issue after attempting self-help. Document each step in the ticket and provide the user with a clear resolution summary.

1. 1. Verify ticket details: confirm user's laptop model, docking station make/model, monitor model, and connection type (DP/HDMI).
2. 2. Remote session: request screen-share via Teams or use Remote Assistance to view Display Settings and device manager.
3. 3. Perform power-cycle and cable swap as described in the self-help procedure, noting any change in behavior.
4. 4. Check dock firmware version via the vendor utility; if out-of-date, schedule firmware update (may require a reboot).
5. 5. Validate graphics driver version; if outdated or corrupted, download and install the latest driver from the manufacturer's website (Dell, Lenovo, HP, Intel, NVIDIA, AMD). Record the pre- and post-update versions.
6. 6. Adjust resolution and refresh rate to the monitor's native specifications; apply changes and ask the user to confirm stability for at least 5 minutes.
7. 7. If flickering persists, test with a known-good monitor and/or cable to isolate the faulty component.
8. 8. Replace defective hardware (cable, dock, or monitor) according to spares availability; update the asset record in the inventory system.
9. 9. Close the ticket with a resolution note summarizing the root cause and the fix applied, and remind the user of the self-help steps for future reference.

Verification and Closure

After the fix, confirm that the external monitor is consistently detected, displays the correct resolution, and does not flicker under normal workload. Ask the user to reproduce their typical workflow (e.g., video conferencing, CAD work) for 10-15 minutes. If the issue does not recur, mark the ticket as resolved and provide the user with the KB article link for future reference.

When to escalate

If the issue persists after completing the self-help procedure and the service desk has verified dock firmware, graphics

drivers, and cables, escalate to Desktop Support L2 (or the appropriate hardware vendor). Include in the escalation: laptop model, docking station model and firmware version, monitor model, EDID logs, current graphics driver version, and a description of any error messages or event log entries observed.

Keywords: external monitor, docking station, display not detected, flickering screen, DisplayPort, HDMI, dock firmware, graphics driver, resolution, refresh rate